COMP233 — Discrete Mathematics

Lecture 3.2

Predicates and Quantified Statements II

Ruwa' AbuHweidi @ 2026

What makes a statement false?

“Every student in COMP233 submitted the assignment.”

ONE counterexample

One enrolled student who did not submit is enough to falsify the statement.

Today: formalize this reasoning using quantifiers, implication, and negation.

Prerequisites: predicates and domains

Predicate

P(x) becomes a proposition once x is assigned an element of a specified domain D.

Universal ∀

∀x ∈ D, P(x) means P(x) is true for every element of D.

Existential ∃

∃x ∈ D, P(x) means at least one element of D satisfies P(x).

Domain matters

∀x ∈ {2,4}, “x is even” is true; over {1,2,4} it is false.

Negation of quantified statements

Universal → existential

$$\neg(\forall x \in D, P(x)) \equiv \exists x \in D, \neg P(x)$$

Existential → universal

$$\neg(\exists x \in D, P(x)) \equiv \forall x \in D, \neg P(x)$$

Negate the predicate AND swap the quantifier. Keep the domain unchanged.

Why “not all” does not mean “none”

Original

All mathematicians wear glasses.

Correct negation

At least one mathematician does not wear glasses.

Incorrect stronger statement

No mathematician wears glasses. (This is not equivalent.)

Worked example: universal statement

Let D = {1, 2, 3, 4} and P(x): “x is a multiple of 3.”

Original

∀x ∈ D, P(x) is FALSE (e.g., P(1) is false).

Negation

∃x ∈ D, ¬P(x) is TRUE. Witness: x = 1.

To refute ∀: exhibit one counterexample.

Worked example: existential statement

Let D = {1, 2, 3} and P(x): “x > 5.”

Original

∃x ∈ D, P(x) is FALSE; no element exceeds 5.

Negation

∀x ∈ D, ¬P(x) is TRUE: each element is at most 5.

To refute ∃: establish that there is no witness.

Universal conditional: the key structure

Statement

$$\forall x \in D, (P(x) \rightarrow Q(x))$$

Negation

$$\exists x \in D, (P(x) \wedge \neg Q(x))$$

A counterexample must satisfy P but fail Q. Merely failing P does not disprove the implication.

Deriving the conditional negation

Step 1 $\neg\forall x(P(x) \to Q(x))$

Step 2 $\equiv \exists x\, \neg(P(x) \to Q(x))$

Step 3 $\equiv \exists x\, (P(x) \wedge \neg Q(x))$

Use the quantifier-negation law, then $\neg(P \to Q) \equiv P \wedge \neg Q$.

Application: software correctness

Claim: Every program that compiles is bug-free.

Predicates

P(x): x compiles successfully. Q(x): x is bug-free.

Refutation

Find a program that compiles AND contains a bug.

Compiling successfully is not, by itself, evidence of correctness.

Quantifiers over finite domains

Suppose D = {a, b, c}.

Universal behaves like AND

$\forall x \in D, P(x) \equiv P(a) \wedge P(b) \wedge P(c)$

Existential behaves like OR

$\exists x \in D, P(x) \equiv P(a) \vee P(b) \vee P(c)$

Example: finite-domain evaluation

$D = \{1, 2, 3\}$; $P(x)$: x is even.

Universal = FALSE

$P(1) \land P(2) \land P(3) = F \land T \land F = F.$

Existential = TRUE

$P(1) \lor P(2) \lor P(3) = F \lor T \lor F = T.$

Vacuous truth: a surprising boundary case

Statement: “All balls in the empty box are blue.”

Negation test

“There exists a ball in the box that is not blue.”

Result

The negation is false: no ball exists. Therefore, the universal statement is TRUE.

Vacuous truth does not require an empty domain

Over real numbers, consider: $\forall x\,(x < 0 \rightarrow x^2 \geq 0)$.

Important distinction

This statement is true, but NOT vacuously true: negative real numbers exist.

Genuine vacuous case

$\forall x \in \mathbb{R}$, $(x^2 < 0 \rightarrow x > 10)$ is vacuously true because $x^2 < 0$ never holds.

Conditional variants

Original	$\forall x\,(P(x) \rightarrow Q(x))$
Converse	$\forall x\,(Q(x) \rightarrow P(x))$
Inverse	$\forall x\,(\neg P(x) \rightarrow \neg Q(x))$
Contrapositive	$\forall x\,(\neg Q(x) \rightarrow \neg P(x))$

Which related statements are equivalent?

Valid equivalence

Original ≡ Contrapositive

Another equivalence

Converse ≡ Inverse

Not generally equivalent

Original is NOT generally equivalent to its converse or inverse.

Counterexample: converse error

Original: Every multiple of 4 is even.

Original: TRUE

$\forall n \in \mathbb{Z}, (4 \mid n \rightarrow 2 \mid n)$.

Converse: FALSE

$\forall n \in \mathbb{Z}, (2 \mid n \rightarrow 4 \mid n)$. Counterexample: $n = 2$.

Necessary, sufficient, and “only if”

P sufficient for Q	$P \rightarrow Q$
P necessary for Q	$Q \rightarrow P$
P only if Q	$P \rightarrow Q$
P if Q	$Q \rightarrow P$

Necessary and sufficient: worked example

Policy: If an account is an administrator, it may view the dashboard.

P → Q

P: account is an administrator. Q: account may view dashboard.

Reasoning

Administrator status is sufficient for access; access is necessary for administrator status.

What do we know—and what do we NOT know?

From P → Q

P is sufficient for Q and Q is necessary for P.

Not licensed

We cannot conclude Q → P. Other users may also have dashboard access.

A biconditional P ↔️ Q requires BOTH directions.

Core concepts & misconceptions

Universal	True only if every domain element satisfies P.
Existential	True if at least one witness satisfies P.
Negation	Switch ∀/∃ and negate the predicate.
Counterexample	Find P(x) true and Q(x) false to refute ∀x(P→Q).

Conceptual connections

Negation ↔ De Morgan

∀ behaves like AND; ∃ behaves like OR over finite domains.

Implication ↔ proof

A universal implication is false precisely when a P ∧ ¬Q witness exists.

Conditional variants

Original ≡ contrapositive; converse ≡ inverse.

Worked cases: valid, invalid, boundary

True universal	$\forall n \in \mathbb{Z}, n^2 \geq 0$.
False universal	$\forall n \in \mathbb{Z}, n > 0$; counterexample $n = 0$.
True existential	$\exists n \in \mathbb{Z}, n^2 = 4$; witness $n = 2$.
False existential	$\exists n \in \mathbb{Z}, n^2 = -1$; no integer witness.

Question 1 / 8

QUESTION

Negate: $\forall x \in D, (Student(x) \rightarrow Passed(x))$.

A. $\exists x \in D, (Student(x) \land \neg Passed(x))$

B. $\forall x \in D, (Student(x) \land \neg Passed(x))$

C. $\exists x \in D, (Student(x) \rightarrow \neg Passed(x))$

Question 2 / 8

QUESTION

Over D = {1,2,3}, which is true for P(x): “x is even”?

A. ∀x P(x) only

B. ∃x P(x) only

C. Both

D. Neither

KNOWLEDGE CHECK

Question 3 / 8

QUESTION

True or false: “Not all students passed” means “No students passed.”

A. True

B. False

Question 4 / 8

QUESTION

Find a counterexample: $\forall n \in \mathbb{Z}$, (n is even → n is a multiple of 4).

A. n = 2

B. n = 4

C. n = 8

Question 5 / 8

QUESTION

What is the converse of $\forall x(P(x) \rightarrow Q(x))$?

A. $\forall x(Q(x) \rightarrow P(x))$

B. $\forall x(\neg Q(x) \rightarrow \neg P(x))$

C. $\exists x(P(x) \wedge \neg Q(x))$

KNOWLEDGE CHECK

Question 6 / 8

QUESTION

If P is necessary for Q, what follows?

A. $P \rightarrow Q$

B. $Q \rightarrow P$

C. $P \leftrightarrow Q$

Question 7 / 8

QUESTION

Justify: $\forall x \in \mathbb{R}, (x^2 < 0 \rightarrow x > 100)$.

A. Vacuously true

B. False: $x = 0$

C. False: $x = -1$

Question 8 / 8

QUESTION

Disprove or prove: $\forall n \in \mathbb{Z}, (3 \mid n \rightarrow 6 \mid n)$.

A. True by divisibility

B. False; n = 3

C. False; n = 6

Answers with reasoning — 1 to 4

Q1 — A	¬∀(P→Q) ≡ ∃(P∧¬Q). B is overly strong; C fails to negate the conditional.
Q2 — B	There is an even element (2), but not all elements are even.
Q3 — False	Not all means at least one exception; it does not mean none.
Q4 — n = 2	The antecedent is true (even) while the consequent is false (not divisible by 4).

Answers with reasoning — 5 to 8

Q5 — A

Converse reverses P and Q. The contrapositive negates and reverses.

Q6 — B

P is necessary for Q means Q cannot hold without P: $Q \rightarrow P$.

Q7 — A

No real number has a negative square; the antecedent is always false.

Q8 — B

$n = 3$ is a counterexample: $3 \mid 3$ but 6 does not divide 3.

Optional enrichment: computing and logic

Software verification

Correctness conditions can be formalized as universal implications; failing tests provide counterexamples.

Knowledge representation

Quantifiers encode general rules and existential claims in first-order logical systems.

Next lecture: nested quantifiers and why their order matters.

References & course resources

Primary text: Susanna S. Epp, Discrete Mathematics with Applications, 4th ed., §3.2 (pp. 108–116).

Original course materials — Prof. Mustafa Jarrar

Updated course resources — COMP233 GitHub

With appreciation for subsequent course-material contributions, including Dr. Radi Jarrar.